Team 047: WeCanThinkAboutIt
Project Title: GradPath: A Career Path Analytics Platform for Students

Summary:
GradPath is a database web application designed to alleviate the uncertainty and stress for students that do not have a clear picture of possible career choices and employment opportunities within their expertise. Many students come into college with the notion that a job is lined up once they graduate, which cannot be further from the truth depending on the competitiveness of the job market. Employment opportunities change all the time and vary drastically from region, current economic level, graduation cycle, and industry trends.
With this in mind, our application is designed to integrate all of these analytics into one platform that produces a transparent view on specific career paths based on user preferences. Users will enter their major, degree level, university, and other criteria which will help our system produce unemployment rate statistics, salary trends, and overall job stability indicators in specific industries/sectors and regions. A student's education is also considered since higher level education is also another option for most students looking to pursue higher paying job roles. This is functionally different from a job board as this platform is not designed to host job opportunities but to help point students and other users towards job titles they should be pursuing and also provide industry analytics based on their major and degree level. The platform is designed with the intention to help students evaluate their current career trajectory and make them reassess if they should pursue different interior paths with the help of GradPath's suggestions and data visualizations.

Creative Component:
Our creative component will be in the form of a complex data visualizations dashboard that serves as an input and update in real time based on user input. An example would be an interactive map that when zoomed in provides real time updated data visualizations tied to that particular location. It provides useful functionality to the user because instead of reading through static information that provides no real translation of what users learn from those statistics, a real time data visualization can help users differentiate which statistics will offer them the most information that is important to them.

Usefulness:
GradPath is useful because it helps students better understand how their academic choices translate into real-world career outcomes. Many students assume that selecting a major guarantees employment opportunities, but job prospects vary significantly depending on economic conditions and graduation timing. Simple features include inputting academic background and career interests. Also viewing employment trends in their field. More advanced features include highlighting how factors affect career prospects and showing personalised results.

While similar information exists across government dashboards, statistical reports, and job boards such as LinkedIn or Indeed, these platforms typically present aggregated or impersonal data. Government sites often provide broad graphs, and job boards focus primarily on listings rather than career outlook. GradPath differentiates itself by transforming large-scale labor statistics into individualized insights. Instead of simply showing trends, it helps students understand what those trends mean for someone with their specific background. Additionally, the platform aims to present this information in a more approachable and supportive way, helping reduce uncertainty and stress surrounding employment outcomes.

Realness:
We are using unemployment data from Kaggle that tracks unemployment in the US from 1976 to 2025. The data is sourced from the Bureau of Labor Statistics. This dataset allows us to incorporate regional employment conditions into career insights and is in CSV format. The record includes attributes like state, year, and unemployment rates.
https://www.kaggle.com/datasets/justin2028/unemployment-in-america-per-us-state

We can also join with salary by major using another kaggle dataset in CSV form, containing over 1000 records so the system can display expected earnings associated with different careers. Includes major, median salary, and unemployment rate.
https://www.kaggle.com/datasets/thedevastator/uncovering-insights-to-college-majors-and-their

We will also use a Kaggle dataset on Higher Education Institutions in the United States, provided in CSV format. This dataset contains information on colleges and universities across the country, including institution name, location, type, and enrollment characteristics. With thousands of institutional records, it enables the system to connect user-entered university information to regional employment and education trends. This allows GradPath to contextualize career outlook data based on where a student is studying and how institutional characteristics relate to broader workforce outcomes.
https://www.kaggle.com/datasets/jacksonbarreto/higher-education-institutions-in-the-usa

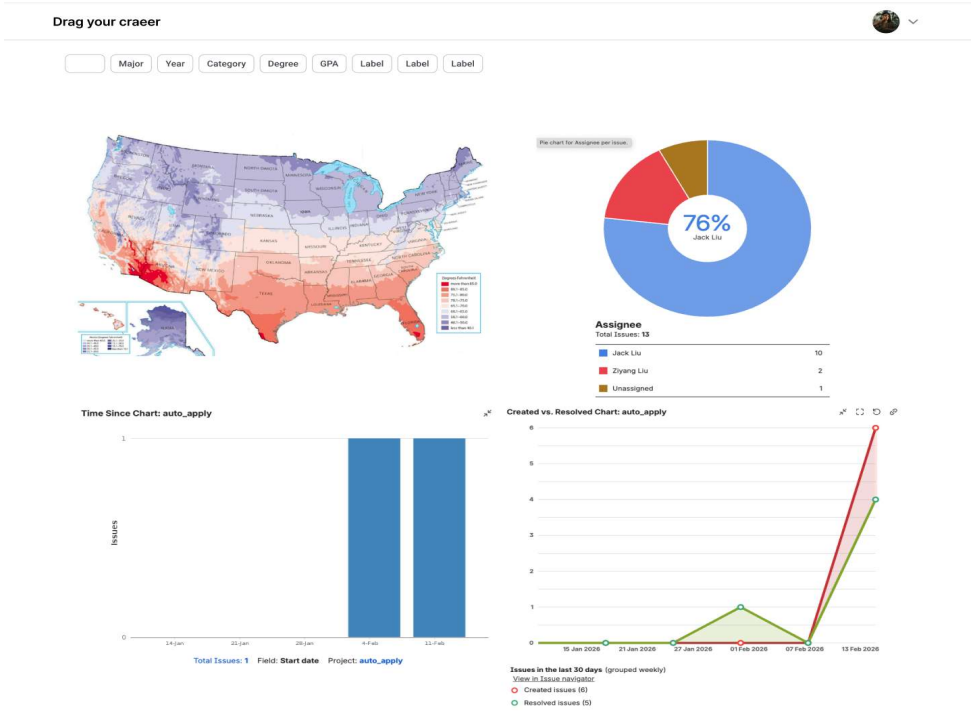
We also are using a Job Placement Dataset from Kaggle, which is also provided in CSV format and contains over 1,000 records. This dataset includes attributes such as education level, specialization, salary, and placement status. It captures how academic background and skill-related factors influence employment outcomes. TThis dataset will help model how education level will impact job placement status and salary expectations.

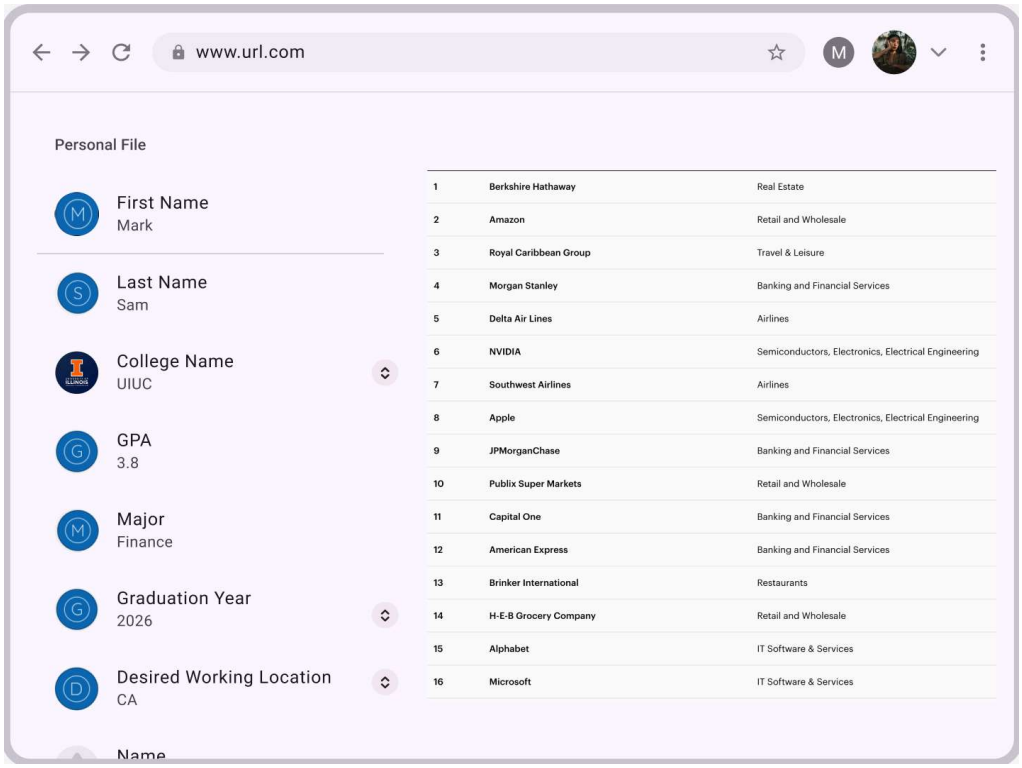
https://www.kaggle.com/datasets/mahad049/job-placement-dataset/data

Functionality:
The application functions as an interactive industry/market analytics and job role matching platform. Users input their university, major, degree level, location which serve as a baseline for a user in the database. Users are allowed to add more details such as industry/sectors they are interested in, salary expectations, graduation year, work interests, etc. These details are stored in user specific tables and are extra parameters/filters for queries pertaining to that information. The system also produces a match score that is assigned to relevant job roles based on the user's preferences and information. This match score is then used to rank relevant job roles by sector/industry and is provided to the user in the form of a list of suggestions for job roles within an industry.

Aside from GradPath's suggestions, users also have a customizable dashboard where they can see different visualizations that pertain to specific data they want to view. For example, the user will have an option to view a heatmap of specific job role popularity based on location and see employment rates in those particular cities. They will also have options to view salary trends, unemployment/employment rates, and job role growth metrics. Users have access to their own generated view/tables allowing them to insert/update/delete any information that is outdated to reflect updated changes on their dashboard as well as be able to favorite particular filter presets. Users will never have access to static databases and datasets as it will tamper with data integrity. Admins however will have access to change particular aspects of the database and insert new datasets if needed.

UI Mockups:


Drag your craeer
Major
Year
Category
Degree
GPA
Label
Label
Label
Pie chart for Assignee per issue.
76%
Jack Liu
Assignee
Total Issues: 13
Jack Liu 10
Ziyang Liu 2
Unassigned 1
Time Since Chart: auto_apply
Issues
1
0
14-Jan
21-Jan
28-Jan
4-Feb
11-Feb
Total Issues: 1 Field: Start date Project: auto_apply
Created vs. Resolved Chart: auto_apply
6
5
4
3
2
1
0
15 Jan 2026
21 Jan 2026
27 Jan 2026
01 Feb 2026
07 Feb 2026
13 Feb 2026
Issues in the last 30 days (grouped weekly)
View in Issue navigator
Created issues (6)
Resolved issues (5)



Work Distribution:

All members are expected to help in areas that have gaps but we will form designated teams for backend tasks:

1. Database design - Yash, Ocean
2. Clean and merging data - Jackie, Jacob
3. Database implementation - ALL Members
4. User Access Controls - Jackie, Jacob
5. Backend APIs - Yash, Ocean
6. QA testing - ALL Members

1->In charge of documentation of ER diagram and defining entities/relationships/keys
2-> Cleaning and merging data will involve assessing datasets for any inconsistent fields, records and data types. They are also in charge of standardizing dataset formats and making data accessible for the database integration.
3->All members are expected to write SQL scripts to create tables, enforce key constraints, and implement stored procedures.
4->Defining user access privileges and making sure CRUD operations to static databases are limited to Admins.
5->Translating queries and aggregations into frontend data visualizations.

6-> All members are expected to test and fix issues related to data integrity, function logic, and any operational issues with the project.